

RESEARCH ARTICLE

Trait-based recovery of insect functional diversity following pine removal in a shrubland biodiversity hotspot

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Abstract

Introduction: A primary goal of habitat restoration is the return of both taxonomic and functional diversity to support ecosystem resilience and functioning. This study assessed how insect functional biodiversity (focusing on beetles [Coleoptera] and ants [Formicidae]) responds to invasive pine (*Pinus* spp.) removal, in a Mediterranean-type ecosystem (the Cape Floristic Region).

Objectives: In Table Mountain National Park, we assessed morphological traits of beetles and ants across four biotopes: active pine plantations, areas cleared 10 years ago (“new recovery”), cleared 30 years ago (“old recovery”), and areas never planted with pine (“pristine fynbos”) to evaluate how functional trait composition responds to passive restoration over time.

Methods: Insects were sampled using a combination of pan traps, pitfall traps, and a vacuum sampler, with sampling conducted over three seasons.

Results: Our results show striking differences in recovery trajectories. Functional diversity of both taxa was lowest in pine plantations. Beetle assemblages showed relatively rapid functional recovery, with diversity in recovery stages comparable to pristine fynbos within a decade. Herbivorous beetles recovered faster than predatory species, suggesting feeding guilds as useful recovery indicators. Among ants, ground-nesting species showed faster functional recovery during early restoration stages, whereas opportunistic nesters recovered more slowly, reflecting differing ecological requirements. Overall, ant functional diversity remained greatest in pristine fynbos and showed limited recovery even after 30 years.

Conclusions: These findings highlight that although beetle functional diversity can recover within one decade, ant assemblages may require considerably longer.

Implications for Practice: Successful long-term restoration of insect functional biodiversity, particularly for ecologically important predatory taxa, requires sustained efforts in removing invasive pine (*Pinus* spp.), promoting native vegetation regrowth, and controlling secondary invasive species. These actions help reestablish key ecosystem functions such as nutrient cycling, herbivory regulation, and seed dispersal. Passive recovery may lead to partial restoration over decades; however, active interventions, such as reseeded native plants, restoring structural vegetation complexity, use of appropriate fire and managing invasive undergrowth, are likely to accelerate the process. Supporting functionally diverse insect communities through such measures enhances ecosystem resilience and ensures the reassembly of trophic interactions.

Key words: biodiversity conservation, insect recovery, Mediterranean ecosystem, passive restoration, Table Mountain National Park, trait expression

Introduction

Insect populations are experiencing dramatic global declines driven primarily by human activities (Hallmann et al. 2017; Seibold et al. 2019; Cardoso et al. 2020). Among the major threats to biodiversity is the spread of alien invasive plants, including non-native pine plantations. Pines (*Pinus* spp.) have been widely introduced across the Southern Hemisphere as a source of timber, creating severe ecological and economic challenges (Richardson & Petit 2006; Pauchard et al. 2015). Several species have become highly invasive, and their impacts have intensified over recent decades. Pine invasions have substantially altered native biodiversity, disrupted hydrological processes, and increased wildfire risks (Richardson & Petit 2006; Van Wilgen & Richardson 2012). In South Africa, formal pine plantations occupy 660,000 ha; in contrast, invasive stands

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occupy an additional 2.9 million hectares (van Wilgen & Richardson 2012). Previous studies have shown that monoculture pine stands support low faunal diversity and offer limited conservation value (Pryke & Samways 2009; Cifuentes-Croquevielle et al. 2020; Cawood et al. 2025). These plantations also support low floral diversity and vegetation cover, resulting in simplified habitat structure with few niches for foraging, shelter, and predator avoidance (Jouveau et al. 2022). This issue is particularly acute in the mountainous regions of the Cape Floristic Region (CFR), where invasive pine stands have caused substantial ecological disruption and are now the focus of ongoing removal efforts (van Wilgen 2015). Clearing and restoration of these areas may allow for the recovery of fynbos plant species, but the response of insects to pine clearing remains poorly understood (Pryke & Samways 2009; Cawood et al. 2025).

Although the ecological importance of insects to ecosystem processes is well established, there remains limited understanding of how insect functional traits, and by extension, functional diversity, respond to land use change and climate pressures (deCastro-Arrazola et al. 2023). Studies that rely exclusively on taxonomic metrics such as species composition and species richness often fall short in predicting ecological responses to disturbance (Elek et al. 2022). In contrast, functional trait-based approaches offer greater insight into the mechanisms underlying community assembly, the resilience of ecosystem functions, species' extinction risk, and how assemblages respond to environmental change (Fountain-Jones et al. 2015; Grab et al. 2019).

Trait-based ecology has progressed rapidly over the past two decades, with the most notable advances occurring in plant ecology, where functional traits have been widely applied to predict community assembly and ecosystem functioning (Moretti et al. 2017; Wong et al. 2019). In contrast, applications of trait-based approaches to terrestrial insects, particularly within the context of ecosystem restoration, remain comparatively limited (Barber et al. 2017; Wong et al. 2019; García-Atencia et al. 2024). Nonetheless, recent initiatives have made strides in compiling trait databases for key insect groups, including carabid beetles (Homburg et al. 2014), saproxylic beetles (Bouget et al. 2008), and ants (Bertelsmeier et al. 2013; Parr et al. 2017). An expanding body of research is beginning to document how functional trait expression in taxa such as ants, bees, beetles, butterflies, and spiders responds to environmental pressures (Schirmel et al. 2012; Wong et al. 2019). However, these relationships remain poorly understood for many other insect groups. Linking trait expression to environmental variables offers a promising pathway to predict how insect communities may respond to global change, thereby helping to identify species that are likely to persist or decline under different management scenarios.

Functional traits are defined as measurable characteristics of an organism, such as morphology, behavior, feeding strategies, or life cycle attributes, that influence its responses to environmental conditions and its effects on ecosystem functioning (Díaz et al. 2016). Trait-based frameworks typically distinguish between two major categories: (1) functional response traits, which determine how species respond to environmental variation and are key to understanding species distributions, and

(2) functional effect traits, which reflect how species influence ecosystem processes, community structure, or environmental conditions, often at broader spatial and temporal scales (Violle et al. 2007). Fountain-Jones et al. (2015) proposed a structured framework for analyzing functional traits in adult beetles, focusing on four trait types: morphological (e.g. body size, color, and wing length), physiological (e.g. food and microclimate preferences), phenological (e.g. breeding season and season of peak abundance), and ecological (e.g. habitat and feeding guild). This M–P–P–E framework provides a basis for linking beetle traits to both environmental drivers and ecosystem functions.

Linking morphological traits to their ecological functions allows a better understanding of how species respond to environmental change and contribute to ecosystem processes (Violle et al. 2007; Díaz et al. 2016). Traits such as body size, dispersal ability, and defensive structures influence reproduction, movement, habitat use, and species interactions (Fountain-Jones et al. 2015). For example, body size is associated with dispersal capacity and habitat filtering (Ness et al. 2004; Ng et al. 2018), while traits like color, pilosity, and cuticle sculpturing relate to thermoregulation and desiccation resistance (Law et al. 2020; Buxton et al. 2021). In restoration contexts, analyzing these traits helps identify which species are likely to persist or return following disturbance, and which functional roles may be lost or recovered (Barnes et al. 2014; Wong et al. 2019). Trait-based analyses can therefore provide deeper insights into community assembly, ecosystem recovery, and resilience beyond what species identity alone can reveal (Fountain-Jones et al. 2015; Grab et al. 2019; Elek et al. 2022).

A wide range of indices exist to quantify functional diversity, yet there is limited consensus regarding what each metric captures, how much redundancy exists among them, and which are most appropriate for specific ecological questions (Mouchet et al. 2010). To address this, we use four complementary indices: functional richness, functional dispersion, Rao's quadratic entropy (RaoQ), and overall functional diversity. Functional diversity broadly refers to the component of biodiversity that influences ecosystem functioning, by capturing the range and value of functional traits within a community (Tilman 2001). Functional richness describes the volume of trait space occupied by a community, reflecting the extent of niche space filled by species (Mouchet et al. 2010). Functional dispersion measures the spread of species in trait space, capturing trait variance and the distribution of species relative to the trait-space centroid (Daniel & Rooney 2022). RaoQ quantifies the mean pairwise dissimilarity in trait values among all species in a community, integrating both trait variation and species abundances (Botta-Dukát 2005). Together, these indices provide various views of functional assemblages and potential ecosystem functioning.

We focused on beetles (Coleoptera) and ants (Formicidae) as focal taxa. Ants are highly sensitive to environmental change and play key ecological roles (e.g. soil turnover, seed dispersal, and predation) (Sithole & Tantsi 2021). They have therefore been used as restoration indicators since the mid-1970s, when ants were first employed in Australia to assess mining impacts. Additionally, combined ant–beetle analyses have proven

valuable for assessing ecosystem recovery (Leach et al. 2013). Beetles, including carabids, cerambycids, and dung beetles, have also been successfully used as bioindicators in various restoration contexts (Maleque et al. 2009). Numerous studies demonstrate that beetle functional traits respond predictably to habitat disturbance and recovery, making them useful indicators of functional ecosystem change (Audino et al. 2014; Evans et al. 2019). Together, ants and beetles offer complementary insights into restoration processes, making them ideal taxa for this study.

This study investigates the functional biodiversity of beetles and ants across four biotope types: (1) pine plantations, (2) new recovery (cleared 10 years ago), (3) old recovery (cleared 30 years ago), and (4) sites never planted, referred to as “pristine fynbos.” The main aim is to assess insect recovery following pine removal by analyzing functional richness, functional diversity, functional dispersion, RaoQ, and trait composition. Specifically, we address three objectives: to (1) determine which recovery stage most closely resembles pristine fynbos in terms of functional assemblages, (2) identify which functional traits respond to environmental changes driven by restoration, and (3) evaluate the potential of beetles and ants as bioindicators of restoration success post pine removal. We hypothesize that old recovery sites will be most functionally similar to pristine fynbos across all functional diversity indices, reflecting progressive recovery over time. We also predict that functional traits associated with dispersal will be more prevalent in recovery sites than in pristine fynbos, reflecting colonization dynamics, while traits linked to stable habitat conditions, such as larger body size, will be more common in pristine fynbos. Additionally, we expect herbivorous species to recolonize more rapidly than other feeding guilds due to increased resource availability from regenerating vegetation. Finally, we anticipate that beetles and ants will demonstrate recovery and that their functional trait responses will validate their use as indicator taxa for ecosystem restoration. This study contributes to the limited understanding of insect functional recovery following pine removal and passive restoration and provides insights to guide management in this globally important biodiversity hotspot.

Methods

Site Selection

We conducted this study in Table Mountain National Park (TMNP) bordered by the City of Cape Town, within the south-western CFR. The CFR ranks among the world's six floral kingdoms (Goldblatt 1997; Manning & Goldblatt 2012) and 35 global biodiversity hotspots, hosting approximately 9400 vascular plant species, 69% of which are endemic (Mittermeier et al. 2011). TMNP's dominant vegetation is fynbos, a fire-prone and fire-adapted shrubland with a Mediterranean-type climate of cool, wet winters and hot, dry summers. Fire records show that none of the sampled sites had burned since the 1960s. Thus, all treatments represent long-unburnt Peninsula Granite Fynbos, with differences reflecting time since pine removal rather than recent post-fire succession.

Following Pryke and Samways (2009) previous sampling coordinates (2005–2007), we selected 32 sites (Fig. 1) along the eastern slopes of TMNP within Peninsula Granite Fynbos, spanning Constantia Nek to Newlands. In Constantia Nek, pine plantations were surrounded by fynbos, while in Newlands, they were bordered by indigenous forest. Although efforts were made to maximize spatial separation, some degree of spatial autocorrelation was unavoidable due to the limited extent of recovering fynbos in Constantia Nek. Additionally, pine plantation sites generally occurred at lower elevations than fynbos sites. The 32 sites were categorized into four biotopes, with eight replicates each, spaced ≥ 400 m apart. These were pine plantations (“pine”), areas cleared approximately 10 years prior (“new recovering fynbos”), areas cleared approximately 30 years prior (“old recovering fynbos”), and never-planted fynbos (“pristine fynbos”).

Insect Sampling

We used three complementary sampling methods to target different insect communities. Pitfall traps captured epigeic insects using 70 mm diameter, 90 mm deep plastic containers, sufficiently wide to capture most invertebrates while minimizing vertebrate bycatch, inserted flush with the soil. Traps used double-container systems for easy removal without soil disturbance. Four traps arranged in $1\text{ m} \times 1\text{ m}$ squares per site were preinstalled with lids for at least 1 week prior to sampling to minimize digging-in effects. During sampling, lids were removed, and traps were half-filled with a 50% ethylene glycol solution and left open for 1 week before contents were processed and preserved in 75% ethanol.

Vacuum sampling collected foliage-dwelling insects using a Stihl SH56C-E vacuum shredder (Stihl, Germany), fitted with a 30 cm-long fine mesh collecting bag attached to the 10 cm diameter nozzle (Swart et al. 2017). This method is deemed both practical and minimally disruptive in fynbos (Pryke & Samways 2008). We standardized sampling by inserting the nozzle into vegetation and moving it vertically through the foliage 40 times per site. Samples were frozen overnight, then transferred to 75% ethanol.

Pan traps sampled aerial insects on calm, sunny days for maximum efficiency. We placed four colored pan traps (yellow, white, red, and blue) at the corners of a $5\text{ m} \times 5\text{ m}$ square in randomized positions, because multiple colors increase sampling effectiveness; yellow and white attract a broad range of species, and red and blue may capture rarer taxa (Vrdoljak & Samways 2012). Traps contained water and a small amount of detergent to reduce surface tension and were left in place for 8 hours, after which their contents were sieved and preserved in 75% ethanol.

Species were identified as morphospecies at least to family level, and genus and species where possible, using identification guides and keys (Scholtz & Holm 1985; Picker et al. 2019). These field guides were also used to identify non-native species.

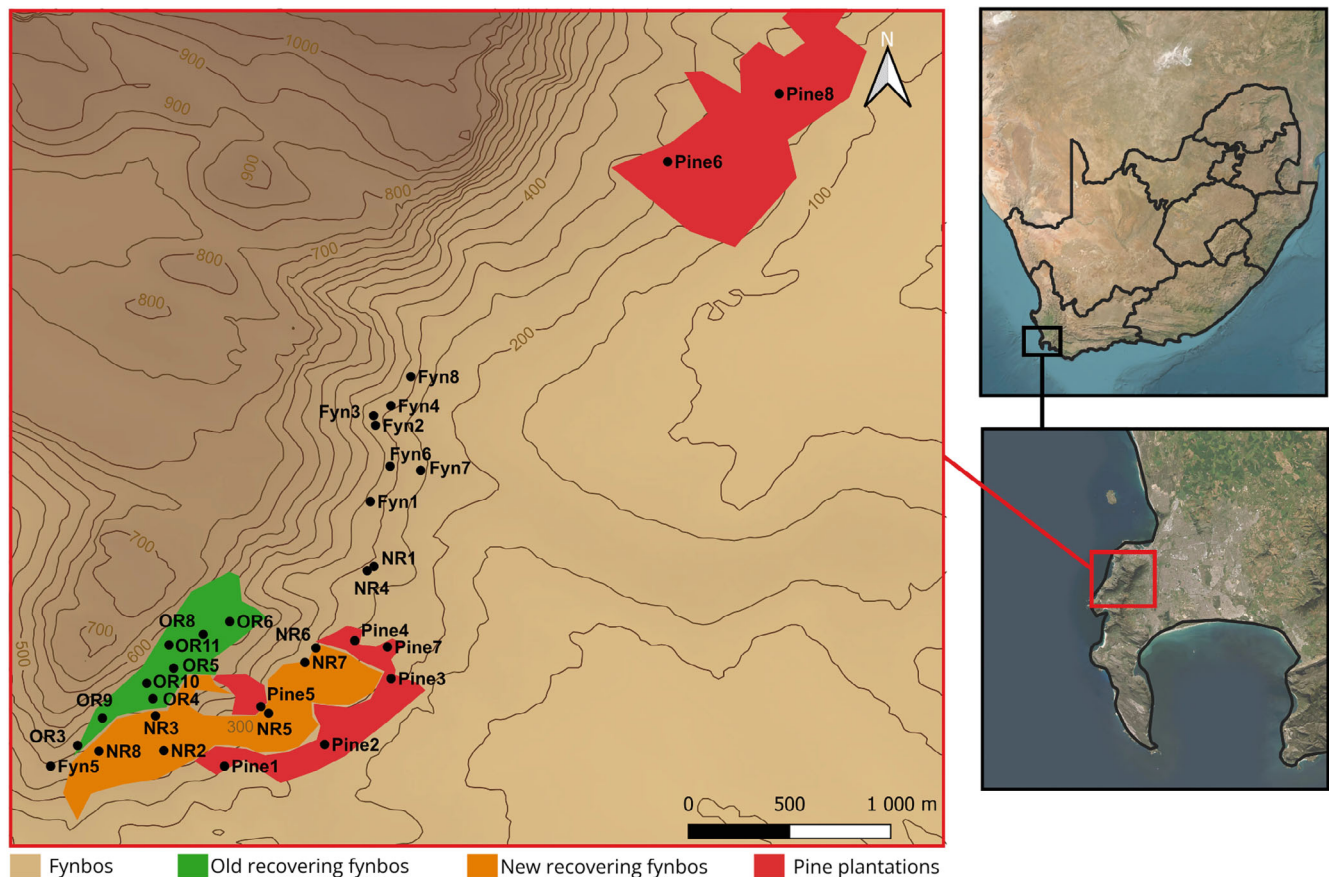


Figure 1. Left: A map showing the study area and the location of sites. Fyn, fynbos; OR, old recovering fynbos; NR, new recovering fynbos; Pine, pine plantations. Top right: study area shown with respect to South Africa. Bottom right: study area with respect to the Cape Peninsula.

Trait Measurements

Fountain-Jones et al. (2015) outline a comprehensive set of beetle traits relevant for assessing ecological responses to habitat change. From this suite, we selected four key traits for beetles: body size, dispersal power (calculated as leg length divided by body length), color variation, and feeding guild (Table S1). For ants, we focused on morphological traits linked to foraging and dispersal: body size, dispersal power (Almeida et al. 2023; Drager et al. 2023) and, given the ecological importance of habitat vertical stratification, nesting type (arboreal, leaf litter, and soil) (Vogel et al. 2024). We included additional morphological ant traits such as color variation, number of spines, pilosity, and cuticle sculpturing, following trait descriptions in the Global Ants Database. Feeding guild was excluded for ants since 20 of 21 species sampled were predators (with some opportunistic sap feeders), with only one herbivorous species found (Table S2). For each species, six individuals were measured and averaged (see Table 1).

Statistical Analysis

For the statistical analysis, season and trap type were pooled. Using the *ggplot2* package (Wickham 2016) in R version 4.5.1 (R Core Team 2024), we visualized trait data. For beetles, we

visualized body length, dispersal ability and color within feeding guilds per biotope as well as the relationship between body length and dispersal ability. For ants, we visualized body length, dispersal ability, color, pilosity, and number of spines within each nesting type per biotope as well as the relationship between body length and dispersal ability.

Using the *HillR* package in R (Li 2018; R Core Team 2024), we calculated functional richness, RaoQ, functional dispersion, and total functional diversity for each site for all beetles, predatory beetles, herbivorous beetles, all ants, ground-nesting ants, and opportunistic ants. Distribution models were tested using *DHARMa* (Hartig 2022), *lme4* (Bates et al. 2015), and *MASS* (Venables and Ripley 2002) packages in R (R Core Team 2024) to identify the appropriate error distribution for generalized linear mixed models (GLMM). GLMMs were performed in R (R Core Team 2024) to determine the effect of biotope on functional richness, RaoQ, functional dispersion and total functional diversity for all beetles, predatory beetles, herbivorous beetles, all ants, ground-nesting ants, and opportunistic ants. This was followed by a Tukey's pairwise test to identify which biotopes differed significantly from each other.

We used the *mvabund* (Wang et al. 2012) and *lattice* (Sarkar 2008) packages in R (R Core Team 2024), to visualize associations between species traits and each biotope using fourth

Table 1. Summary of traits measured for beetles and ants and their possible functional links. Trait type: E, ecological; M, morphological.

Trait	Trait type		Unit	Measurement	Possible functional links
Body length	M	mm		Tip of head to end of abdomen	Generation time Microhabitat use Foraging capabilities
Dispersal	M	Ratio		Top of femur to end of tarsus of hind leg	Dispersal Colonization Foraging capabilities Hunting Escape from predators
Color	M	Six colors: black, dark brown, light brown, gold, yellow, and metallic green		Dominant color of elytra in beetles and cuticle in ants	Thermoregulation Aposematism Camouflage
Feeding guild	E	Four guilds: predators, herbivores, detritivores, and nectarivores		Species feeding preference	Resource use
Nesting type	E	Three nest types: arboreal, ground, and opportunistic		Species nesting preference	Microhabitat use
Pilosity	M	Numerical		Number of hairs on the profile of the mesosoma	Thermoregulation Desiccation resistance
Spines	M	Numerical		Number of spines on the mesosoma	Defense mechanism Foraging capabilities
Sculpturing	M	Three categories: smooth cuticle (1), shallow wrinkles (2), and heavily textured (3)		Texture of cuticle	Thermoregulation

corner analyses. This was done for all beetles, predatory beetles, herbivorous beetles, all ants, ground-nesting ants and opportunistic ants.

Results

There were 45 (24 herbivorous, 12 predatory, six nectarivorous, and three detritivorous) species of beetles with six or more individuals for trait measurement (954 individuals in total of which 270 individuals were measured). There were 21 (11 ground-nesting, eight opportunistic, and two arboreal-nesting) species of ants that had six or more individuals for trait measurement (2268 individuals of which 126 were measured).

Trait Visualization Per Biotope

Herbivorous beetles were generally smaller than predators (Fig. S1). Pine plantations lacked nectarivores and detritivores, unlike recovering and pristine fynbos, which supported diverse feeding guilds, including small detritivores and a range of herbivores and predators. Dispersal potential varied among guilds, with detritivores generally showing high potential dispersal ability, herbivores spanning a broad range from low to high, while predators were consistently intermediate (Fig. S2). Pine beetle communities were dominated by intermediate-dispersing herbivores, while predators were rare and high-dispersing predators were absent from new recovery sites but present in older

recovery and pristine fynbos. Beetle body size did not correlate with dispersal (Fig. 2A). Black was the dominant color among beetles except for detritivores, which were dark brown. Recovering and pristine sites had darker predators and a mix of light and dark herbivores (Fig. S3).

Ants showed similar body size and dispersal patterns across biotopes, with small arboreal species and various sized ground-nesting species, including one large species in older recovery and pristine fynbos (Fig. S4). Opportunistic ants were mostly small. Dispersal was medium to high for ground-nesting and opportunistic ants, intermediate for arboreal (Fig. S5). No body size–dispersal correlation was found (Fig. 2B). Ant colors were mainly black and dark brown, with light brown more common in ground-nesters (Fig. S6). Arboreal ants were hairless; ground-nesting and opportunistic ants varied in hairiness, with fewer hairy ants in pine plantations (Fig. S7). Spine numbers were consistent across biotopes, mostly two spines in ground-nesters and opportunists, and two or four in arboreal ants (Fig. S8).

Functional Richness, Diversity, Dispersion, and RaoQ

Functional richness of all beetle species ($\chi^2 = 21.11$, $p < 0.001$, degrees of freedom [df] = 3) was significantly lower in pine than in the other three biotypes (Fig. 3A). Functional diversity of all beetle species ($\chi^2 = 8.95$, $p < 0.05$, $df = 3$) only differed significantly between pine and new recovering fynbos, being significantly higher in new recovering fynbos (Fig. 3B).

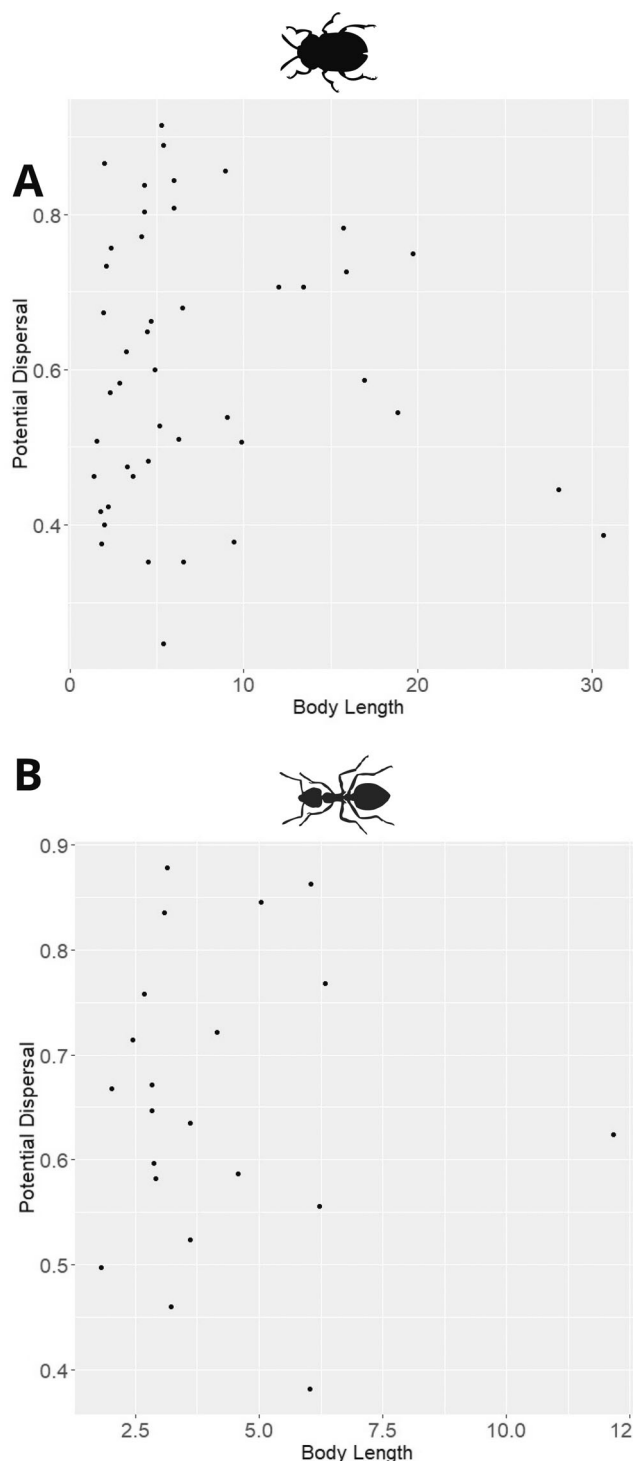


Figure 2. Body length versus potential dispersal ability for (A): beetles and (B): ants. Icons sourced from www.canva.com.

Functional dispersion and RaoQ of all beetle species did not differ significantly between any of the biotopes.

Functional richness, functional dispersion, and RaoQ of predatory beetle species were all significantly lower in pine than in pristine fynbos ($\chi^2 = 9.38$, $p < 0.05$, $df = 3$; $\chi^2 = 10.20$,

$p < 0.05$, $df = 3$; $\chi^2 = 10.66$, $p < 0.05$, $df = 3$; Fig. 4A–C). Functional diversity showed no significant differences among biotopes.

Functional richness of herbivorous beetle species ($\chi^2 = 10.58$, $p < 0.05$, $df = 3$) differed significantly between pine and both new and old recovering fynbos and was lowest in pine (Fig. 4D). In pine, RaoQ of herbivorous beetle species ($\chi^2 = 10.42$, $p < 0.05$, $df = 3$) was significantly lower than that in recovering fynbos (Fig. 4E). Functional diversity and functional dispersion of herbivorous beetles did not differ among biotopes.

Functional dispersion of all ant species ($\chi^2 = 16.19$, $p < 0.01$, $df = 3$) was significantly higher in pristine fynbos than in old recovering fynbos (Fig. 5A). RaoQ of all ant species ($\chi^2 = 11.47$, $p < 0.01$, $df = 3$) was significantly higher in fynbos than in all other biotopes (Fig. 5B). Functional richness and functional diversity of all ant species did not differ significantly among the biotopes.

Functional richness of ground-nesting ant species ($\chi^2 = 10.70$, $p < 0.05$, $df = 3$) was higher in recovering fynbos than in pine (Fig. 5C). Functional dispersion and RaoQ of ground-nesting ant species ($\chi^2 = 12.87$, $p < 0.01$, $df = 3$; $\chi^2 = 11.68$, $p < 0.01$, $df = 3$) were higher in pristine fynbos than in all other biotopes (Fig. 5D & 5E). Functional diversity of ground-nesting ant species did not differ significantly between any of the biotopes.

Functional diversity of opportunistic ant species ($\chi^2 = 9.63$, $p < 0.05$, $df = 3$) was significantly higher in fynbos than in new recovery (Fig. 5F). Functional richness, functional dispersion, and RaoQ of opportunistic ant species did not differ significantly between any of the biotopes.

Trait Associations to Biotopes

Here we report the significant associations with the interaction coefficient in brackets. The association between all beetle species traits and biotopes shows significant positive associations between light brown and new recovering fynbos (0.17) and black and pine (0.13). Weaker positive associations were seen between dispersal and pine (0.06) and between yellow coloration and pine (0.05). There was a significant negative association between dispersal and fynbos (−0.08) and weaker negative associations between black coloration and old recovering fynbos (−0.05) and body length and new recovering fynbos (−0.01) (Fig. 6A). Herbivorous beetles showed light brown colored species associated with new recovering fynbos (0.18), and black species with pine (0.09), plus a weak negative association between black and new recovering fynbos (−0.02) (Fig. 6B). Predatory beetles showed no biotope associations.

Ant traits showed significant positive associations between body length and fynbos (0.49) and four mesosomal spines and new recovering fynbos (0.33). There were some weaker positive associations between new recovering fynbos and black (0.21), old recovering fynbos and dispersal (0.18), and pine and dark brown (0.18). Significant negative associations occurred between new recovering fynbos and textured cuticles (−0.37) and dispersal and pine (−0.41). There were some weaker negative associations between fynbos and black (−0.18), old recovering fynbos and having a textured cuticle (−0.19), old

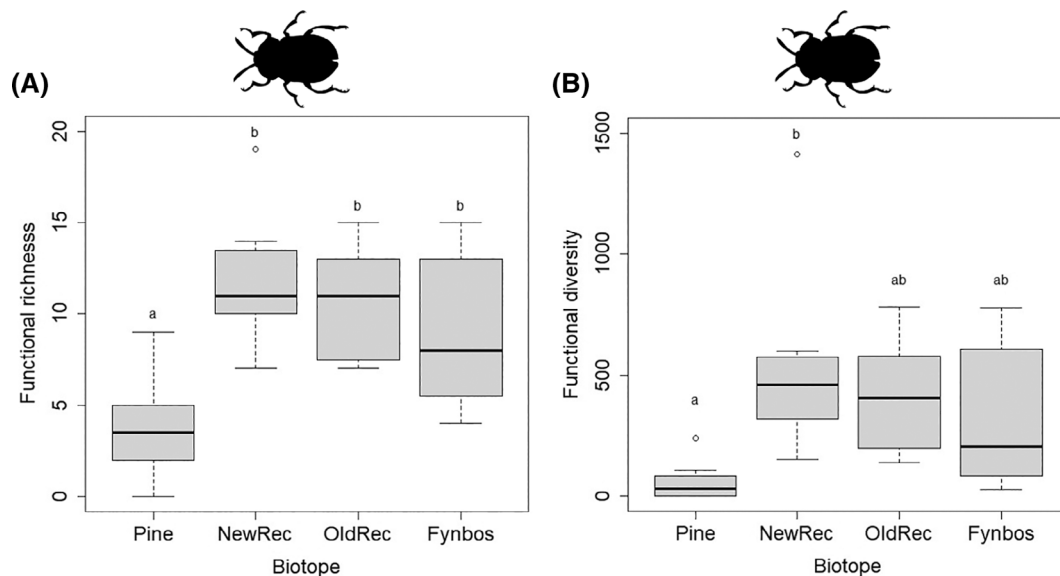


Figure 3. (A) functional richness of all beetles, (B) functional diversity of all beetles. Icons sourced from www.canva.com.

recovering fynbos and four mesosomal spines (-0.15), pine and body length (-0.14), and pine and four mesosomal spines (-0.18) (Fig. 6C).

For ground-nesting ant species body length was significantly positively associated with fynbos (0.78) as well as having a textured cuticle (0.74). Some weaker positive associations were present between old recovering fynbos and having two mesosomal spines (0.59) and new recovering fynbos and black coloration (0.56). There were some negative associations between fynbos and pilosity (-0.43), fynbos and having two mesosomal spines (-0.58) and old recovering fynbos and light brown coloration (-0.35). There were also weak negative associations between fynbos and dispersal (-0.26) as well as pine and dispersal (-0.24) (Fig. 6D).

Body length in opportunistic ant species was significantly positively associated with fynbos (0.81), as was the case for four mesosomal spines (0.72). In pine, species with textured cuticles were more represented (0.57). There were some significant negative associations between fynbos and pilosity (-1.05), fynbos and no mesosomal spines (-0.48), new recovering fynbos and four mesosomal spines (-0.69) and pine and pilosity (-0.75) (Fig. 6E).

Discussion

Invasive pine removal initiates measurable ecological recovery processes, but these vary among insect groups and functional guilds. Different taxa exhibit distinct recovery rates, with some groups, such as beetles, showing relatively rapid taxonomic and functional restoration, while others, such as ants, recover more slowly. Understanding these differential recovery trajectories is important because they highlight where recovery is rapid and largely passive, and where

restoration may remain incomplete without targeted interventions. Such insights can guide more effective management by identifying which components of biodiversity require additional support and which may recover naturally.

Beetle Functional Recovery

Beetle assemblages were functionally impoverished within pine plantations, indicated by significantly lower functional richness compared to recovering and pristine fynbos. Notably, the absence of nectarivores and detritivores in pine sites indicates a loss of key ecological guilds that are essential for pollination and nutrient cycling (Schowalter et al. 2018; Crespo-Pérez et al. 2020; Verma et al. 2023). This functional loss suggests that pine plantations create conditions unsuitable for many beetle guilds, consistent with previous studies showing that such plantations typically support lower faunal diversity and reduced ecological complexity (Pryke & Samways 2009; Corley et al. 2018; Cifuentes-Croquevielle et al. 2020; Cawood et al. 2025).

Within a decade, there was evidence of functional recovery in beetles, with new recovering fynbos having significantly higher functional diversity and richness than pine plantations. These results align with previous findings that beetle assemblages can recover taxonomically within 10 years of pine removal (Cawood et al. 2025), suggesting that both species composition and ecological function can return relatively quickly. Functional richness and diversity remained relatively stable from new recovery through to older recovery and pristine fynbos, pointing to a relatively rapid sustained functional rebound. However, some metrics such as functional dispersion and RaoQ did not differ significantly among fynbos biotopes. This may reflect rapid recolonization by generalist species that restore broad trait coverage, or the persistence of functionally redundant taxa that

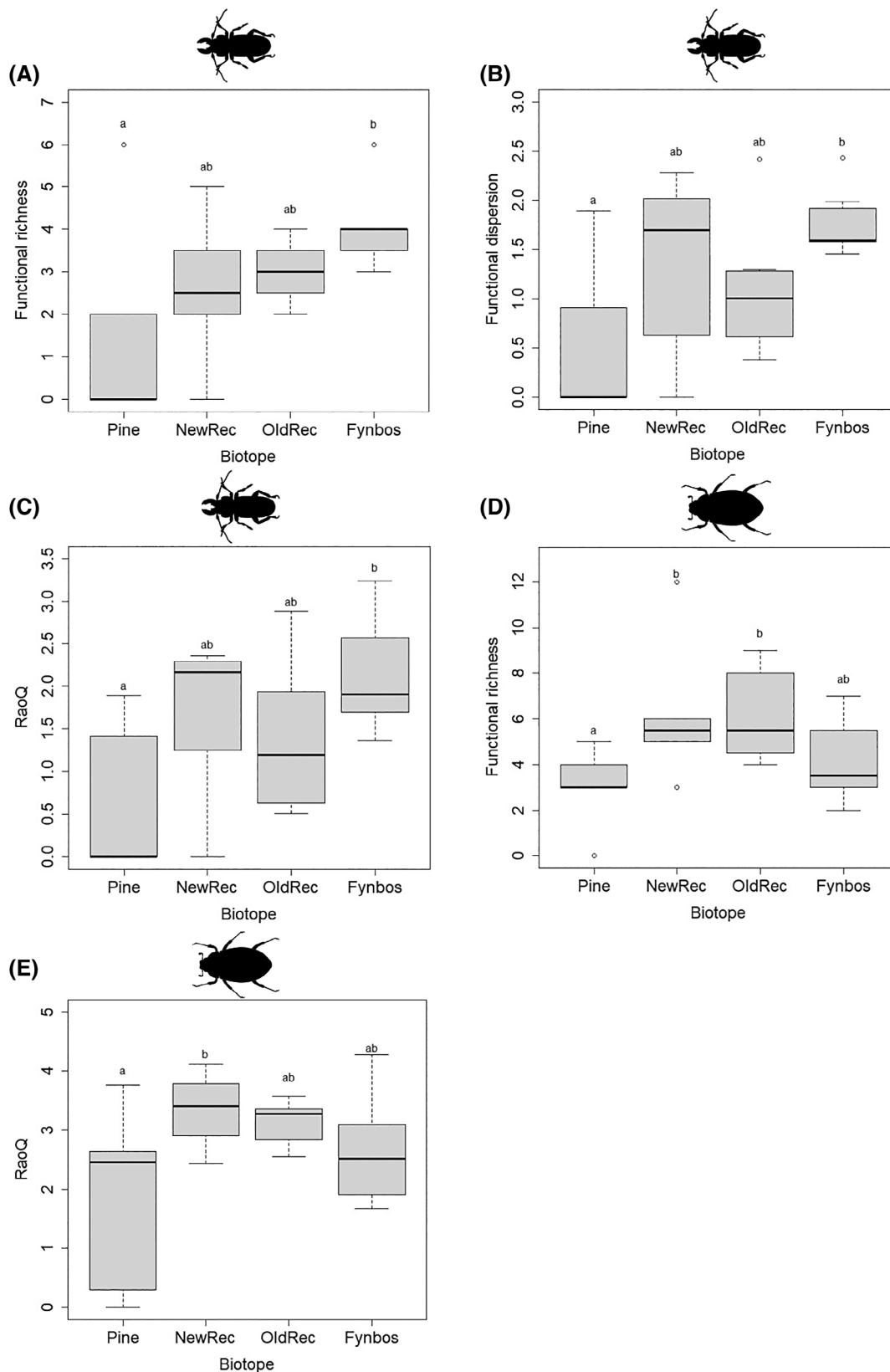


Figure 4. (A) functional richness of predatory beetles, (B) functional dispersion of predatory beetles, (C) RaoQ of predatory beetles, (D) functional richness of herbivorous beetles, and (E) RaoQ of herbivorous beetles. Icons sourced from www.canva.com.

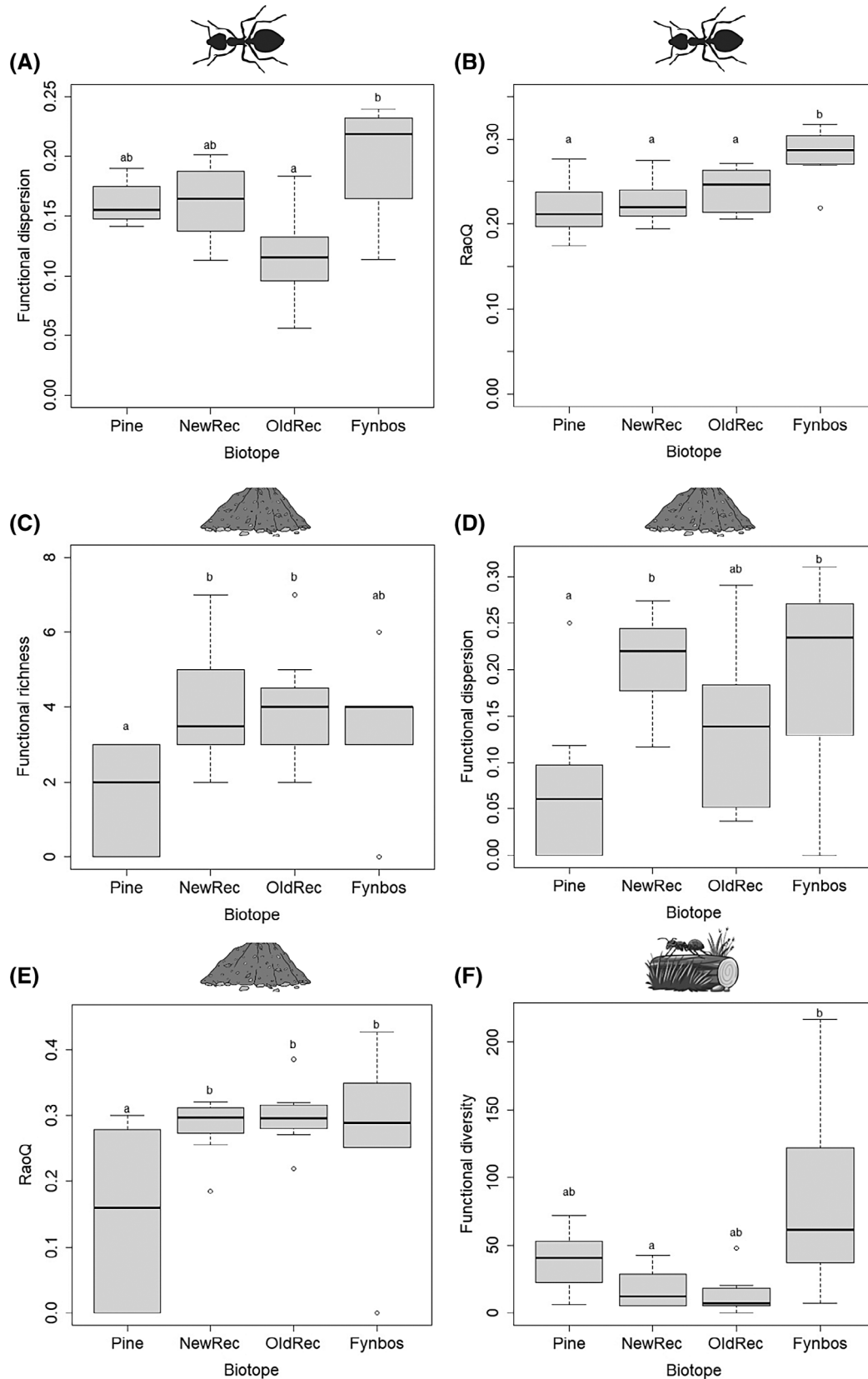


Figure 5. (A) Functional dispersion of all ants, (B) RaoQ of all ants, (C) functional richness of ground-nesting ants, (D) functional dispersion of ground-nesting ants, (E) RaoQ of ground-nesting ants, and (F) functional diversity of opportunistic ants. Icons sourced from www.canva.com.

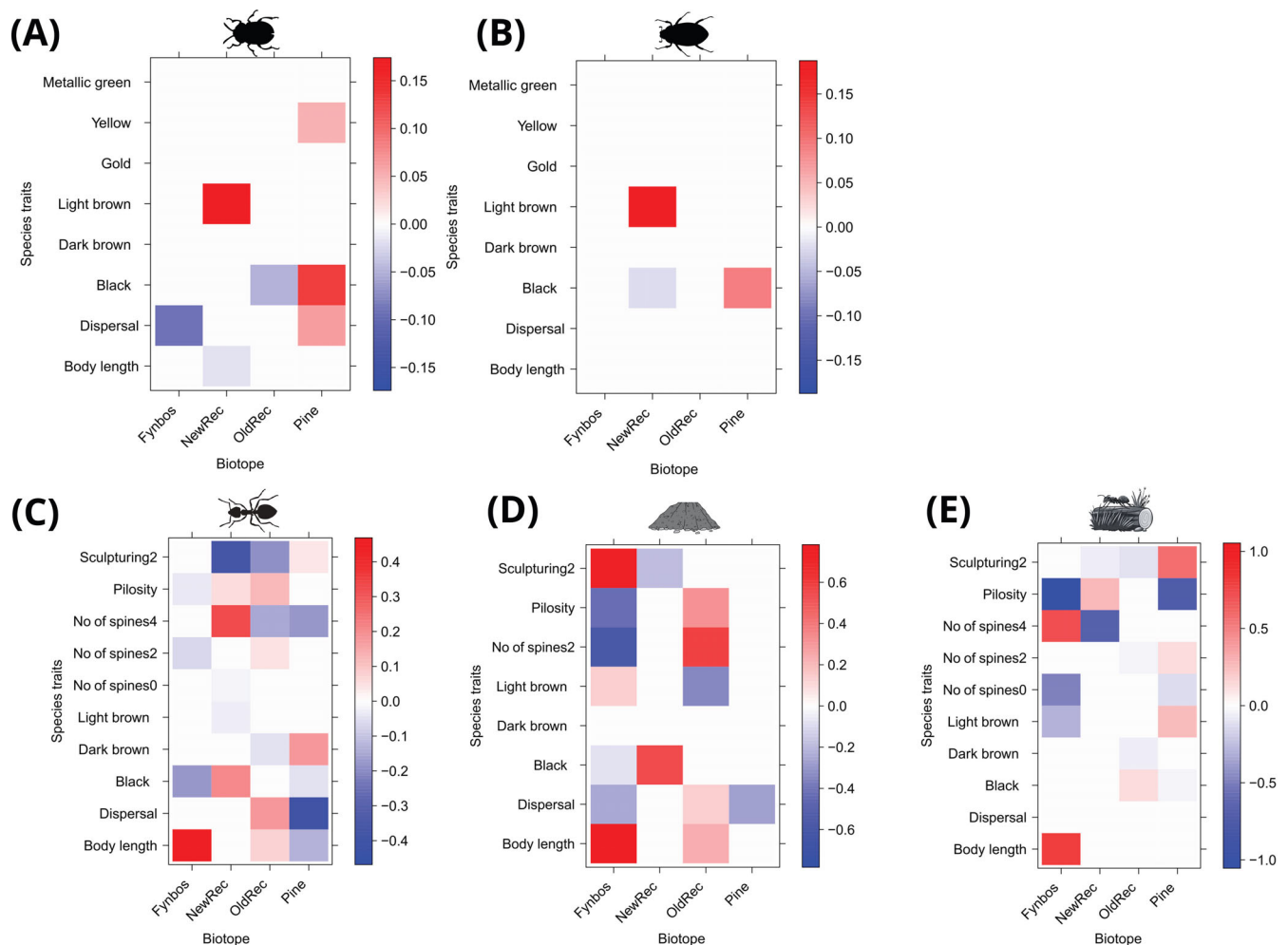


Figure 6. Fourth corner plot: Showing interaction coefficients between species traits (y-axis) and biotope (x-axis) for (A) all beetle species, (B) herbivorous beetle species, (C) all ant species, (D) ground-nesting ant species, and (E) opportunistic ant species. Significant positive associations are red and negative associations are blue. Icons sourced from www.canva.com.

maintain trait structure even when more specialized functions have yet to recover fully (Woodcock et al. 2014).

Predatory beetles exhibited lower functional richness and diversity in pine plantations relative to pristine fynbos, with recovery sites showing intermediate values. This suggests partial functional recovery, where generalist predators may recolonize relatively early, but more specialized taxa remain absent. The lack of significant differences between recovery and either extreme (pine or pristine) implies a transitional state, pointing to a longer recovery time frame for predatory beetle assemblages, likely influenced by prey availability, habitat complexity, and species-specific dispersal constraints (Liu et al. 2014). In contrast, herbivorous beetles showed significantly higher functional richness and RaoQ in new recovering fynbos compared to pine plantations, with these metrics remaining relatively stable through older recovery and pristine stages. This indicates that herbivorous beetles respond more rapidly to vegetation recovery, likely because they rely on plant resources that become available early in succession. Herbivores appear

earlier than predators in ecological succession, consistent with previous studies showing their preference for young leaves and increased food availability (Choi et al. 2010). Flowering native plants often recover after pine removal, which likely increases food resources for herbivorous insects. Together, these findings underscore differing restoration trajectories among functional groups, with specialist or higher trophic-level species requiring longer time frames or more structurally complex habitats for full reestablishment (van Nouhuys 2005; Liu et al. 2014).

Beetle trait expression revealed ecologically meaningful shifts along the recovery gradient. Multiple ecological filters, including habitat complexity, resource distribution, and inter-specific interactions are likely to influence beetle recolonization during recovery (e.g. Lassau et al. 2005). The observed negative association between dispersal ability and pristine fynbos indicates that high-dispersal generalists dominate recovering sites, while low-dispersal, resident species persist in intact habitats. Additionally, the positive association between high-dispersal

traits and pine plantations further supports the idea that disturbed or simplified habitats favor mobile generalist species, while pristine fynbos sustains more resident, low-dispersal specialists. Color traits also varied by biotope: light brown beetles were more common in recovering fynbos, whereas black beetles dominated pine plantations. These patterns likely reflect adaptations to microclimatic conditions and vegetation complexity, where lighter morphs may benefit from sun-exposed environments or camouflage in open vegetation (Zverev et al. 2018). Together, these trait shifts demonstrate that beetle communities respond to both broad habitat changes and fine-scale environmental filters during recovery.

Ant Functional Recovery

Ant assemblages exhibited a slower and more gradual recovery compared to beetles, with functional diversity metrics indicating prolonged sensitivity to past pine cover. Functional dispersion and RaoQ were significantly higher only in pristine fynbos, suggesting that functional recovery in ants occurs over extended timescales. This pattern aligns with previous taxonomic findings (Cawood et al. 2025), which showed that ant communities remain distinct from pristine fynbos even 30 years post-clearing. Pine plantations are characterized by poor understorey vegetation and homogenous litter, limiting nesting, foraging, and microclimatic buffering (Pacheco et al. 2009). While recovering fynbos supports a high diversity of generalist pioneer plants, it still lacks the structural complexity and specialized features of pristine vegetation, potentially constraining full functional recovery in ant communities.

Ground-nesting ants exhibited clear signs of functional recovery in the early stages of restoration, with significantly higher functional richness, dispersion, and RaoQ in new recovering fynbos compared to pine plantations. These metrics remained relatively stable through older recovery stages and pristine fynbos, suggesting that some components of the ant community can respond relatively quickly to vegetation recovery. This is likely due to their reliance on soil and microhabitat conditions, which begin to improve as native plant communities reestablish (Rocha-Ortega & García-Martínez 2018). These results indicate that while overall ant community recovery is slow, certain functional groups such as ground-nesters may exhibit earlier signs of functional response to restoration.

Opportunistic ants exhibited higher functional diversity in pristine fynbos compared to recovering sites, while functional richness and dispersion were relatively unchanged across biotopes. This pattern suggests that the full recovery of trait complexity is slower and likely tied to the progressive development of microhabitat heterogeneity and ecological interactions, such as competition, mutualisms, and colony dynamics, that shape functional roles over longer timescales (Rocha-Ortega & García-Martínez 2018).

Ant trait–environment relationships further support patterns of delayed functional recovery. Larger ants were more common in pristine fynbos, likely due to their ability to navigate structurally complex habitats (Farji-Brener et al. 2004). Traits such as textured cuticles were associated with mature fynbos, reflecting

adaptations to stable microhabitats and environmental stresses like desiccation and predation (Buxton et al. 2021). In contrast, ants in newly recovering sites exhibited early successional traits, such as darker coloration and higher spine counts, possibly reflecting responses to open, warmer conditions and increased foraging competition (Blanchard et al. 2020; Law et al. 2020). Surprisingly, textured cuticles were less common in recovering stages despite their potential advantages. Dispersal traits were weakly associated with old recovery and negatively associated with pine plantations, suggesting that harsh or disconnected pine environments constrain colonization. No relationship was found between body size and dispersal, contrary to previous findings (Ness et al. 2004; Gómez & Espadaler 2013). Overall, trait patterns reflect a gradual filtering of ant communities by increasing habitat complexity and ecological interactions over time.

Overall Functional Recovery

Together, these findings highlight that restoring insect functional diversity after invasive pine removal is a gradual process, with certain functional traits and guilds recovering more rapidly than others. Beetle communities, particularly predatory guilds, exhibit delayed recovery, likely due to complex trophic needs and dispersal limitations. Ant communities show some early functional trait recovery but still depend on mature habitat characteristics to regain full functional complexity.

This underscores the importance of long-term monitoring of both taxonomic and functional diversity to effectively evaluate restoration success. Restoration efforts should focus not only on increasing species richness but also on enhancing habitat features that support a broad range of functional traits vital for ecosystem processes.

Pine plantations are also important ecological analogs for pine invasions across the CFR, where invasive trees form dense stands that suppress native vegetation (Van Wilgen et al. 2022). Clearing plantations provides insight not only into post-plantation restoration, but the recovery of invaded fynbos landscapes. However, vegetation recovery following pine removal is highly variable and depends on invasion intensity, duration, and regional context, with some sites recovering relatively rapidly while others show strong legacy effects of invasion (Holmes et al. 2020). Importantly, these trajectories should not be interpreted as a simple linear process of recovery over time, but rather as recovery patterns shaped within the context of extended fire absence. This complexity suggests that passive restoration may not always be sufficient, and that active interventions such as seeding or planting of native species, and perhaps even controlled burning, may be required in heavily invaded or long-invaded sites.

Our results show that insect functional diversity recovers gradually after pine removal, but whether full recovery can be achieved through passive processes alone remains uncertain. For plants, Holmes et al. (2020) raised this question, and a similar debate is needed for insects. Some guilds with strong dispersal abilities (e.g. flying pollinators and aerial predators) may be able to recolonize where passive restoration is used, but others with limited dispersal or specialized habitat

requirements (e.g. detritivores, especially terrestrial crustaceans and springtails) may lag as they do post-fire (Pryke & Samways 2012).

Although active reintroduction of insects has been attempted for some taxa elsewhere (e.g. butterflies in Europe), it is rarely feasible in diverse systems like the fynbos. Many fynbos insects are highly host-specific and are part of complex ecological interactions; thus successful reintroductions would require fully restored habitat conditions and are therefore unlikely to be feasible at scale. Recovery of insect assemblages in fynbos will likely depend largely on passive recolonization from nearby source populations. The key management question is therefore how to maximize connectivity, reduce dispersal barriers, and accelerate vegetation recovery to support natural insect return.

Importantly, if vegetation recovers but insect communities remain impoverished, ecosystem functioning may still be compromised. Many ecosystem processes in fynbos, such as pollination, herbivory, decomposition, and soil turnover, are tightly linked to insect diversity. This highlights the need to consider both faunal and floral recovery in restoration planning, and to develop indicators that capture the joint recovery of plants and their associated insect assemblages. Future research should investigate the specific factors restricting trait recovery, such as microhabitat quality, biotic interactions, and landscape connectivity, to better inform restoration strategies. Integrating functional trait analyses with traditional diversity assessments offers a powerful approach to understanding and guiding ecosystem recovery in fire-prone biodiversity hotspots like the fynbos.

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Supporting Information

The following information may be found in the online version of this article:

- Figure S1.** Body length of beetle species per feeding group in (A) pine, (B) new recovering fynbos, (C) old recovering fynbos, and (D) fynbos.
- Figure S2.** Dispersal of beetle species in (A) pine, (B) new recovering fynbos, (C) old recovering fynbos, and (D) fynbos.
- Figure S3.** Color of beetle species in (A) pine, (B) new recovering fynbos, (C) old recovering fynbos, and (D) fynbos.
- Figure S4.** Body length of ant species per nesting type in (A) pine, (B) new recovering fynbos, (C) old recovering fynbos, and (D) fynbos.
- Figure S5.** Dispersal of ant species per nesting type in (A) pine, (B) new recovering, (C) old recovering, and (D) fynbos.
- Figure S6.** Color of ant species in (A) pine, (B) new recovering, (C) old recovering, and (D) fynbos.
- Figure S7.** Pilosity of ant species per nesting type in (A) pine, (B) new recovering fynbos, (C) old recovering fynbos, and (D) fynbos.
- Figure S8.** Number of spines in ant species per nesting type in (A) pine, (B) new recovering fynbos, (C) old recovering fynbos, and (D) fynbos.
- Table S1.** Beetle morphospecies codes and the corresponding species traits.
- Table S2.** Ant species codes and their corresponding trait values.

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